Veille informatique quantique - 2026-11-06

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Machine learning quantum data to accelerate quantum Monte Carlo

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.08151v1 Announce Type: new Abstract: Simulating strongly correlated electrons in molecules and materials remains a central goal for quantum computing. A promising strategy uses quantum processors to prepare trial wavefunctions that guide classical quantum Monte Carlo (QMC). However, the enormous computational cost of repeatedly processing quantum measurements has severely limited the practical scalability of this approach. Here, we overcome this bottleneck by compressing quantum meas

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum processor, quantum algorithm.

2. QDiff: Differential Testing of Quantum Software Stacks

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.08220v1 Announce Type: new Abstract: Over the past few years, several quantum software stacks (QSS) have been developed in response to rapid hardware advances in quantum computing. A QSS includes a quantum programming language, an optimizing compiler that translates a quantum algorithm written in a high-level language into quantum gate instructions, a quantum simulator that emulates these instructions on a classical device, and a software controller that sends analog signals to a ver

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Mots-clés détectés : quantum computing, qubit, quantum algorithm.

3. Hybrid Oscillator-Qubit Quantum Processors: Simulating Fermions, Bosons, and Gauge Fields

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2409.03747v2 Announce Type: replace Abstract: Key to solving many societally-relevant problems in materials science, quantum chemistry and nuclear physics is understanding their fundamental microscopic quantum behaviour. All of these problems can be phrased as fermionic (e.g. electrons) and bosonic particles (e.g. lattice vibrations, photons) interacting with each other. However, solving these systems of strongly-interacting fermions and bosons is out of reach for classical computers and

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Mots-clés détectés : quantum computer, qubit, quantum processor.

4. High-performance cellular automaton decoders for quantum repetition and toric code

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2604.21866v2 Announce Type: replace Abstract: Execution of quantum algorithms on large-scale quantum computers will require extremely low logical error rates and scalable decoding architectures. Local decoders are promising candidates for this task because they avoid the communication and data-processing bottlenecks inherent in global decoding strategies. Here, we present SCALA (Signaling CA with Local Attraction), a non-hierarchical cellular automaton decoder for quantum repetition and t

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, quantum error correction, quantum algorithm.

5. FNO-Based Inverse Co-Design of Semiconducting Quantum-Dot Devices

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.06994v1 Announce Type: new Abstract: Gate-defined semiconductor quantum dots are promising building blocks for scalable spin-based quantum processors, however their operational properties depend on several fabrication and control parameters, such as lithographic gate layout and applied voltages. Existing device architectures have been refined through repeated fabrication and experimental testing, producing layouts that support reliable device operation. However, this process does not

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, quantum processor.

6. Scalable Transpilation for Overcoming Restricted Connectivity in Distributed Superconducting Quantum Architectures

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.07175v1 Announce Type: new Abstract: Distributed Quantum Computing (DQC) offers a promising pathway to scale superconducting quantum hardware, but restricted inter- and intra-module connectivity creates a major transpilation bottleneck. Current DQC transpilation methods often assume idealized hardware connectivity, ignore subsequent intra-module routing, or suffer from substantial computational overheads that limit their applicability at scale. To address this, we introduce a scalabl

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit.

7. Monopoles, Scattering and Generalized Symmetries on a Quantum Computer

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.07960v1 Announce Type: new Abstract: Quantum computers have advanced to the stage that interesting many-body quantum phenomena can be observed. A striking example is the magnetic monopole paradox, a problem that was only solved recently using the language of generalized symmetries. Here we demonstrate a quantum spin chain analogue of magnetic monopole scattering on Quantinuum System Model H2. We implement a novel protocol for preparing a localized quasiparticle wavepacket on top of t

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, Quantinuum.

8. Autler-townes energy-level engineering for high-quality all-microwave qubit gate

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.07967v1 Announce Type: new Abstract: High-fidelity entangling gates are essential for realizing large-scale fault-tolerant superconducting quantum computing, yet existing approaches are limited: frequency-tuning techniques induce decoherence, while all-microwave-driven schemes suffer the trade-off between gate speed and leakage rate. Here, we introduce an energy-level engineering technique that employs the Autler-Townes effect to manipulate the nonlinearity of qubits without perturbi

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit.

9. Crosstalk-Induced Row Hammer Attacks on Quantum Random Access Codes

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.07983v1 Announce Type: new Abstract: Quantum processors remain susceptible to crosstalk-induced effects despite the continuous evolution of their underlying hardware architectures. In this work, we experimentally investigate the effectiveness of crosstalk-induced attacks across six IBM superconducting quantum processors and one IQM processor. Our results demonstrate that such attacks can be launched across all the platforms, due to persistance of crosstalk, although their effectivene

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, quantum processor.

10. Non-Unitary Quantum Butterfly Effect: Theory and Experiment

Source : arXiv Quantum Physics - publié le 2026-11-06

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2611.08152v1 Announce Type: new Abstract: We present a generalization of the out of time order correlator (OTOC) for non-unitary systems. For a Haar-random brickwork qudit-circuit, peppered with local, projective measurement, we calculate the infinite-temperature, ensemble-average OTOC. Utilizing transfer matrix machinery, we find that the OTOC asymptotically decays exponentially and that the decay depends on the rate of measurement. In the thermodynamic limit, with sparse measurement, th

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Mots-clés détectés : quantum computer, Quantinuum.

Méthode

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